

Social Networks — Final Cheat Sheet

SE4095 · The highest-value information from the complete syllabus, compressed for last-minute review. Full explanations live in Parts 1–3 and the Master Revision.

Definitions, one line each

SNA: study of relationships/connections between entities, not just the entities themselves.

Homophily / Heterophily: similar nodes connect / dissimilar nodes connect.

Degree d_i : count of edges touching node i ; in/out for directed graphs.

Walk/Trail/Path: repeats OK / edges distinct / edges+nodes distinct.

Connected/Strong/Weak: path exists / directed path both ways / connected ignoring direction.

Diameter: longest shortest path in the graph.

Bridge / Local bridge: removal disconnects graph / no common neighbor, removal pushes distance >2 (has a span).

Strong Triadic Closure: 2 strong ties with no tie between them = violation.

Tree: no cycles, $|V|=|E|+1$, exactly one path between any pair.

MST: spanning tree with minimum total edge weight.

Planar: drawable with zero edge crossings.

Bipartite / Affiliation: 2 node sets, edges only across / bipartite of people×groups.

Transitivity: friend of my friend is my friend.

Reciprocity: mutual 2-node tie.

Balance: signed triangle stable iff sign product is positive.

Status: directed balance — consistent ranking around a cycle.

Structural equivalence: similar = shares the same neighbors.

Community: ≥ 2 nodes, shared interest + interaction; dense inside, sparse outside.

Explicit/Implicit community: subscribed / emerges from behavior alone.

Clique: subgraph where everyone connects to everyone.

k-plex/k-core/k-shell: almost-clique / min-degree- k subgraph / layer between k - and $(k+1)$ -core.

Modularity Q : actual within-community edges – random-expected within-community edges.

Purity: fraction matching community's majority label (gameable by singletons).

NMI: bounded $[0,1]$ information-theoretic agreement with true labels.

Evolution 6: growth / contraction / merge / split / birth / death.

Formula sheet

Degree

$$C_d = d_i, \text{ norm: } d_i/(n-1)$$

Eigenvector

$$\lambda C_e = A^T C_e, \text{ largest } \lambda$$

Katz

$$C_{Katz} = \beta(I - \alpha A^T)^{-1} \mathbf{1}, \alpha < 1/\lambda_{\max}$$

Betweenness

$$C_b = \sum \sigma_{st}(v_i) / \sigma_{st}; \text{ Brandes } O(nm)$$

Closeness

$$C_c = 1 / \left(\frac{1}{n-1} \sum l_{i,j} \right)$$

Reciprocity

$$R = \frac{1}{m} \text{Tr}(A^2)$$

Jaccard / Cosine

$$\frac{|N_i \cap N_j|}{|N_i \cup N_j|} / \frac{|N_i \cap N_j|}{\sqrt{|N_i||N_j|}}$$

Expected overlap

$$d_i d_j / n$$

Modularity

$$Q = \frac{1}{2m} \sum (A_{ij} - \frac{k_i k_j}{2m}) \delta(c_i, c_j)$$

P / R / F

$$\frac{TP}{TP+FP}, \frac{TP}{TP+FN}, \frac{2PR}{P+R}$$

Degree sum

$$\sum d_i = 2|E|; \text{ odd-degree count is even}$$

Balance test

multiply 3 edge signs; + = balanced

The centrality 6, side by side

Measure	Key idea	Limitation	Clue
Degree	Count direct ties	Blind to rest of network	"most friends"
Eigenvector	Weight ties by neighbor score	=0 possible (directed/DAG)	"connected to important people"
Katz	+flat baseline β	Over-credits celebrities	"no node stuck at zero"
PageRank	Split score across out-links	Needs dangling-node fix	"not everyone a celeb follows is famous"
Betweenness	Fraction of shortest paths through node	Expensive $O(nm)$	"connects two groups"
Closeness	Inverse avg. distance	Breaks if disconnected	"reaches everyone fast"

Concept Selection Guide (condensed)

Says...	Use...
<i>every edge once</i>	Euler's parity check
<i>shortest path, unweighted</i>	BFS
<i>shortest path, weighted</i>	Dijkstra's
<i>connect all, cheapest</i>	Prim's (MST)
<i>friend/enemy triangle stability</i>	Balance theory
<i>who outranks whom</i>	Status theory
<i>same neighbors?</i>	Structural equiv. (Jaccard/Cosine)
<i>popular nodes, "really" similar?</i>	Similarity significance ($d_i d_j / n$)
<i>all members directly connected</i>	Clique/k-plex/k-core
<i>node in >1 community</i>	CPM
<i>max internal vs external density</i>	Modularity Max. (Louvain)
<i>nested/multi-level groups</i>	Hierarchical / Girvan–Newman
<i>probabilistic hidden structure</i>	SBM
<i>huge + fast-changing</i>	Label Propagation
<i>where flow gets stuck</i>	InfoMap
<i>have ground truth</i>	Precision/Recall/F/Purity/NMI
<i>no ground truth</i>	Semantic coherence + modularity/conductance
<i>soft/probabilistic membership</i>	GMM
<i>some labels known</i>	GCN (semi-supervised)

Common misconceptions — final check

Homophily $\neq$ "friends are similar" — it means similarity *drives* tie formation.

Degree centrality convention (in/out/total) always changes the ranking — state which you're using.

Eigenvector centrality can be exactly 0 on directed/DAG graphs — that's *why* Katz exists.

Higher degree $\neq$ higher betweenness/closeness — they can diverge sharply; the divergence is often the real question.

"Balanced" $\neq$ "all positive" — two negatives + one positive is also balanced.

Purity = 1.0 is trivially gamed by singleton communities — never report it alone.

Community detection $\neq$ ordinary clustering — it operates on linked graph structure, not a distance matrix.

Structural equivalence = shared *connections*, not shared attributes — don't conflate with homophily.

MST (Prim's) solves "connect everyone cheaply," not "shortest path $A \rightarrow B$ " — that's Dijkstra's.

Diameter = *longest* shortest path, not the average.